

The Three Questions

Immanuel Kant, *Critique of Pure Reason* (1781)

Carlos Cotrini

What the five weekends behind us did

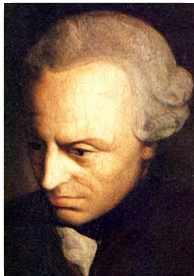
Weekend 0	NumPy, scikit-learn, PyTorch	the tools
Weekend 1	entropy, KL, variance, ensembles	how to measure what you do not know
Weekend 2	confounding, selection, attribution	why a model that scores well can be wrong
Weekend 3	attacks, jailbreaks, privacy	that somebody is trying to break it
Weekend 4	policy gradients, actor-critic, PPO	how a machine chooses under uncertainty

What the five weekends behind us did

Weekend 0	NumPy, scikit-learn, PyTorch	the tools
Weekend 1	entropy, KL, variance, ensembles	how to measure what you do not know
Weekend 2	confounding, selection, attribution	why a model that scores well can be wrong
Weekend 3	attacks, jailbreaks, privacy	that somebody is trying to break it
Weekend 4	policy gradients, actor-critic, PPO	how a machine chooses under uncertainty

None of them let it look anything up, do anything, or run at a size anyone would pay for.

All the interest of my reason



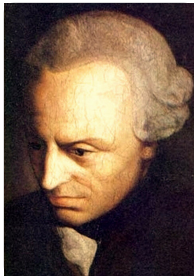
Immanuel Kant (c. 1790),
painter unknown

Alles Interesse meiner Vernunft (das spekulative sowohl, als das praktische)
vereinigt sich in folgenden drei Fragen:

- 1.
- 2.
- 3.

Kritik der reinen Vernunft, A805 / B833; translation C. Cotrini

All the interest of my reason



Immanuel Kant (c. 1790),
painter unknown

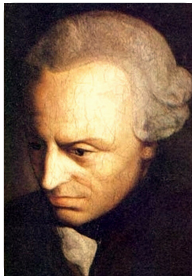
Alles Interesse meiner Vernunft (das spekulative sowohl, als das praktische) vereinigt sich in folgenden drei Fragen:

1. Was kann **ich** wissen?
- 2.
- 3.

What can I know?

Kritik der reinen Vernunft, A805 / B833; translation C. Cotrini

All the interest of my reason



Immanuel Kant (c. 1790),
painter unknown

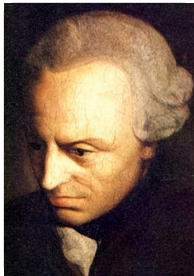
Alles Interesse meiner Vernunft (das spekulative sowohl, als das praktische) vereinigt sich in folgenden drei Fragen:

1. Was kann **ich** wissen?
2. Was soll **ich** tun?
- 3.

What can I know? What ought I to do?

Kritik der reinen Vernunft, A805 / B833; translation C. Cotrini

All the interest of my reason



Immanuel Kant (c. 1790),
painter unknown

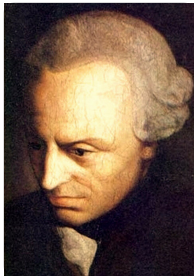
Alles Interesse meiner Vernunft (das spekulative sowohl, als das praktische)
vereinigt sich in folgenden drei Fragen:

1. Was kann **ich** wissen?
2. Was soll **ich** tun?
3. Was darf **ich** hoffen?

What can I know? What ought I to do? What may I hope?

Kritik der reinen Vernunft, A805 / B833; translation C. Cotrini

All the interest of my reason



Immanuel Kant (c. 1790),
painter unknown

Alles Interesse meiner Vernunft (das spekulative sowohl, als das praktische)
vereinigt sich in folgenden drei Fragen:

1. Was kann **ich** wissen?
2. Was soll **ich** tun?
3. Was darf **ich** hoffen?

What can I know? What ought I to do? What may I hope?

Kritik der reinen Vernunft, A805 / B833; translation C. Cotrini

Kant asked these of himself. We will ask them of the machine.

The same three questions, asked of the machine

→

→

→

The same three questions, asked of the machine

Was kann es wissen? → Weekend 5: RAG what it may know beyond its weights

→

→

The same three questions, asked of the machine

Was kann es wissen? → Weekend 5: RAG what it may know beyond its weights

Was soll es tun? → Weekend 6: Agentic AI what it may do in the world

→

The same three questions, asked of the machine

Was kann es wissen? → Weekend 5: RAG what it may know beyond its weights

Was soll es tun? → Weekend 6: Agentic AI what it may do in the world

Was darf es hoffen? → Weekend 7: Large Scale AI what it may become, for what we spend

The same three questions, asked of the machine

Was kann es wissen? → Weekend 5: RAG what it may know beyond its weights

Was soll es tun? → Weekend 6: Agentic AI what it may do in the world

Was darf es hoffen? → Weekend 7: Large Scale AI what it may become, for what we spend

darf, not *wird*: not what will happen, but what it is **entitled to expect**, once we have done our part.

wenn ich nun tue, was ich soll, was darf ich alsdann hoffen?

A scaling law is a warrant for that expectation. It is not a theorem.

The same three questions, asked of the machine

Was kann es wissen? → Weekend 5: RAG what it may know beyond its weights

Was soll es tun? → Weekend 6: Agentic AI what it may do in the world

Was darf es hoffen? → Weekend 7: Large Scale AI what it may become, for what we spend

darf, not *wird*: not what will happen, but what it is *entitled to expect*, once we have done our part.

wenn ich nun tue, was ich soll, was darf ich alsdann hoffen?

A scaling law is a warrant for that expectation. It is not a theorem.

Weekends 1 to 4 were about a model. Weekends 5 to 7 are about a system.